

E2E-TESTS MONITORING:

Practical Insights from the OBI Sales-Rep-Tool

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20.12.23

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Agenda

Characteristics & Key Issues

Sales Rep
Tool

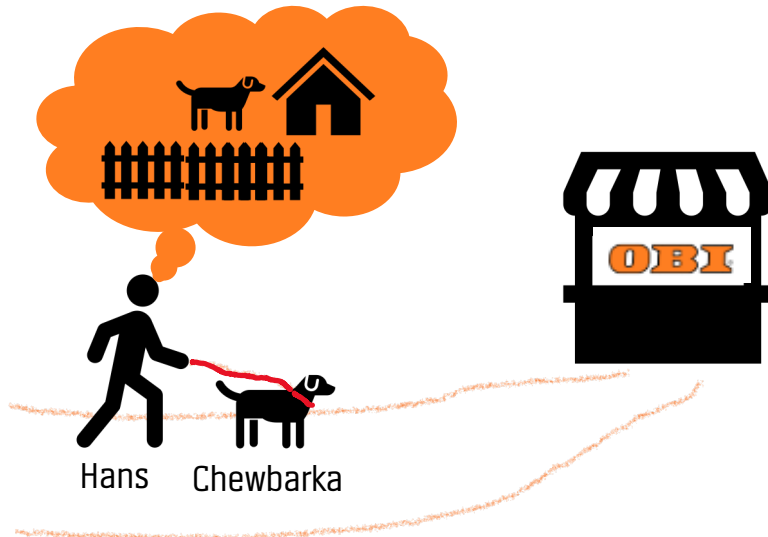
Requirements

Health
check with
E2E tests

How and Why?

E2E test
monitoring
solution

What is Sales-Rep-Tool (germ. „Beratungscenter“)?



What is Sales-Rep-Tool (germ. „Beratungscenter“)?

= a digital tool for the store employees

Sales-Rep-Tool

-  Search customer data
-  Schedule (customer) appointments
-  Manage customer inquiries
-  Communicate with customers
-  Initiate offers
-  Create orders
-  Organize Tasks
-  Take part in training programs

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What is Sales-Rep-Tool (germ. „Beratungcenter“)?

The screenshot shows the OBI Beratungcenter web application. At the top is a navigation bar with the OBI logo and the title 'Beratungcenter'. Below the logo are several icons for navigation: 'Aufgaben' (Tasks), 'Anfragen' (Requests), 'Kennzahlen' (Key Figures), 'Sortiment' (Assortment), 'Kunden' (Customers), and 'Kalender' (Calendar). To the right of these icons is a '+ Neue A' button. Below the navigation bar is a section titled 'Anfragen' (Requests). Underneath this is a section titled 'Offene Angebote und Aufträge' (Open Offers and Orders). This section contains three cards: 'Offene Angebote' (Open Offers) showing 10 offers worth 250,07 €, 'Offene Aufträge' (Open Orders) showing 12 orders worth 16.400,02 €, and 'Abgeschlossen' (Completed) for November 2021 showing 0 orders worth 0,00 €. Below these cards is a section titled 'Neu 1' (New 1). This section has three buttons: 'Überfällig 1 +' (Overdue 1 +), 'Neue Nachricht 1 +' (New Message 1 +), and 'Termin Info' (Appointment Info). Below these buttons is a table with columns: 'To-Do', 'Eingegangen' (Received), 'Name', 'Thema' (Topic), and 'Termin Info'. The table has one row with the date '09.12.2021', the name 'Conrad, Daniel', and the topic 'Dusche'. At the bottom left of the table is the text 'Einträge 1 - 1 von 1'.

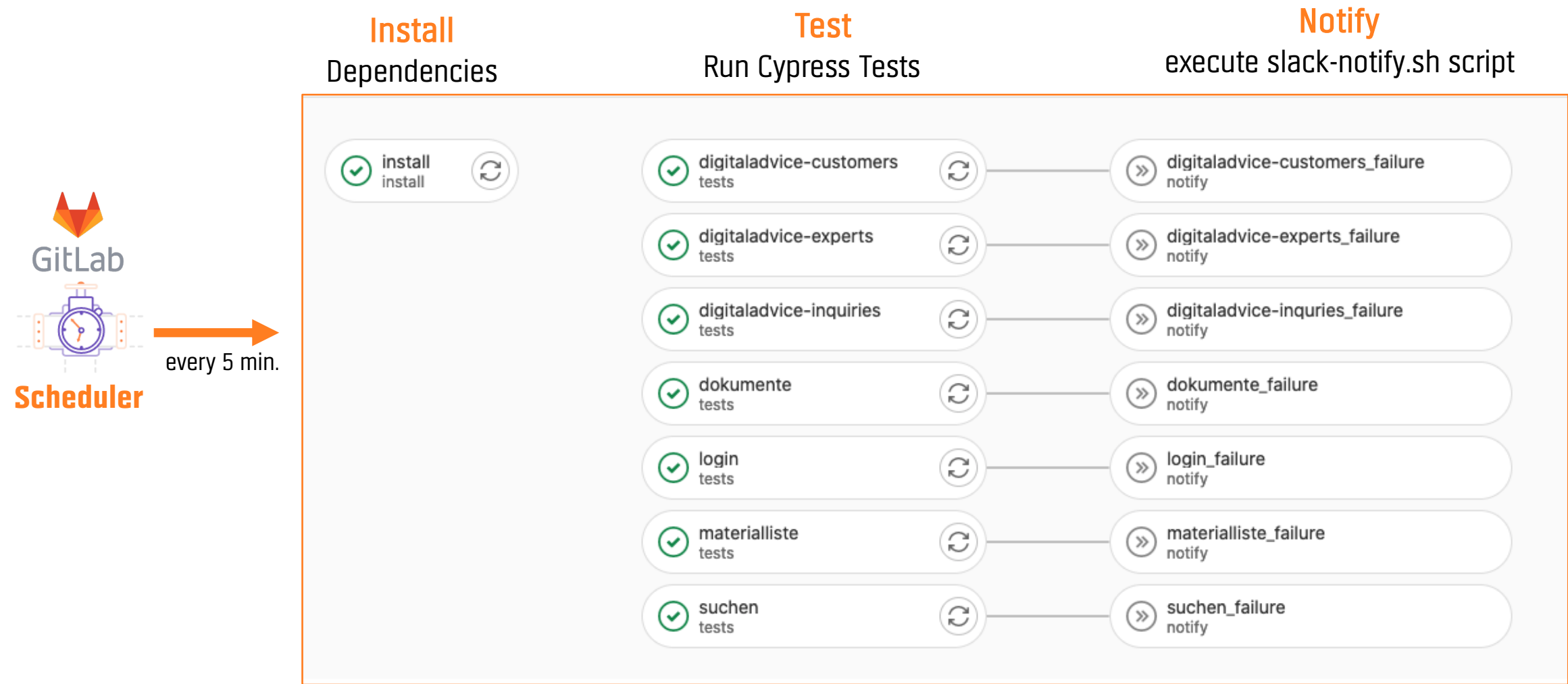
- Microfrontend approach
- Multiple teams
- Independent deployments of FE & BE multiple times a day

Key issues:

- Make sure all parts are working correctly in production
 - The dev team must know about system issues before the users face them
- Continuous E2E testing

Building a Test Monitoring System: The first Attempt

Own Repository for E2E tests



Encountered Problems

- GitLab access required to see the test artefacts like Videos and Screenshots
- Overload of the GitLab pipeline (few available runners for other services)
- Pipeline specific problems like full runner caches → no test execution

! No automated way of knowing whether the tests are still being executed, accept E-Mail for the project team members

Other Issues and Requirements

- Automatic Alerts and Notifications in Slack
- Access to the test results without additional registrational overhead (e.g. VPN)
- Differentiation between test execution errors and application failures
- Alert fatigue – reduce the number of false positive alerts
- Cost efficiency – avoid dependance on paid tools and cloud services like e.g. Cypress Cloud
- Storing historical monitoring data like test statistics and results

So, we asked ourselves ...



How do you monitor the health of your software?

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Software and Tools already used for the Sales-Rep-Tool

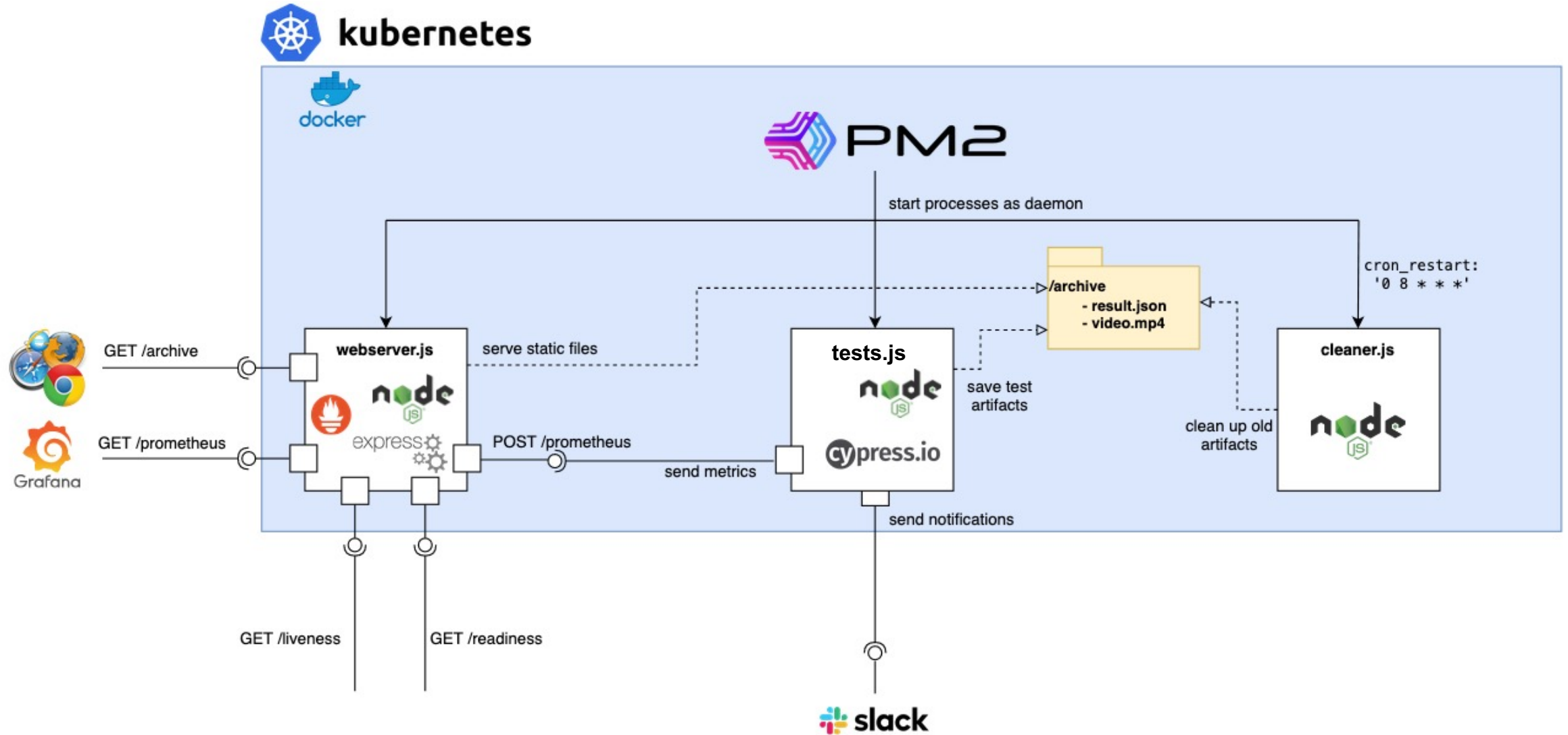
- Used Monitoring Tools:
 - Prometheus for customized metrics
 - Kibana for loggings (now Grafana Cloud Loki)
 - Grafana for Monitoring and Alertings
- Other Tools:
 - Docker and Kubernetes

**OUR
SOLUTION**

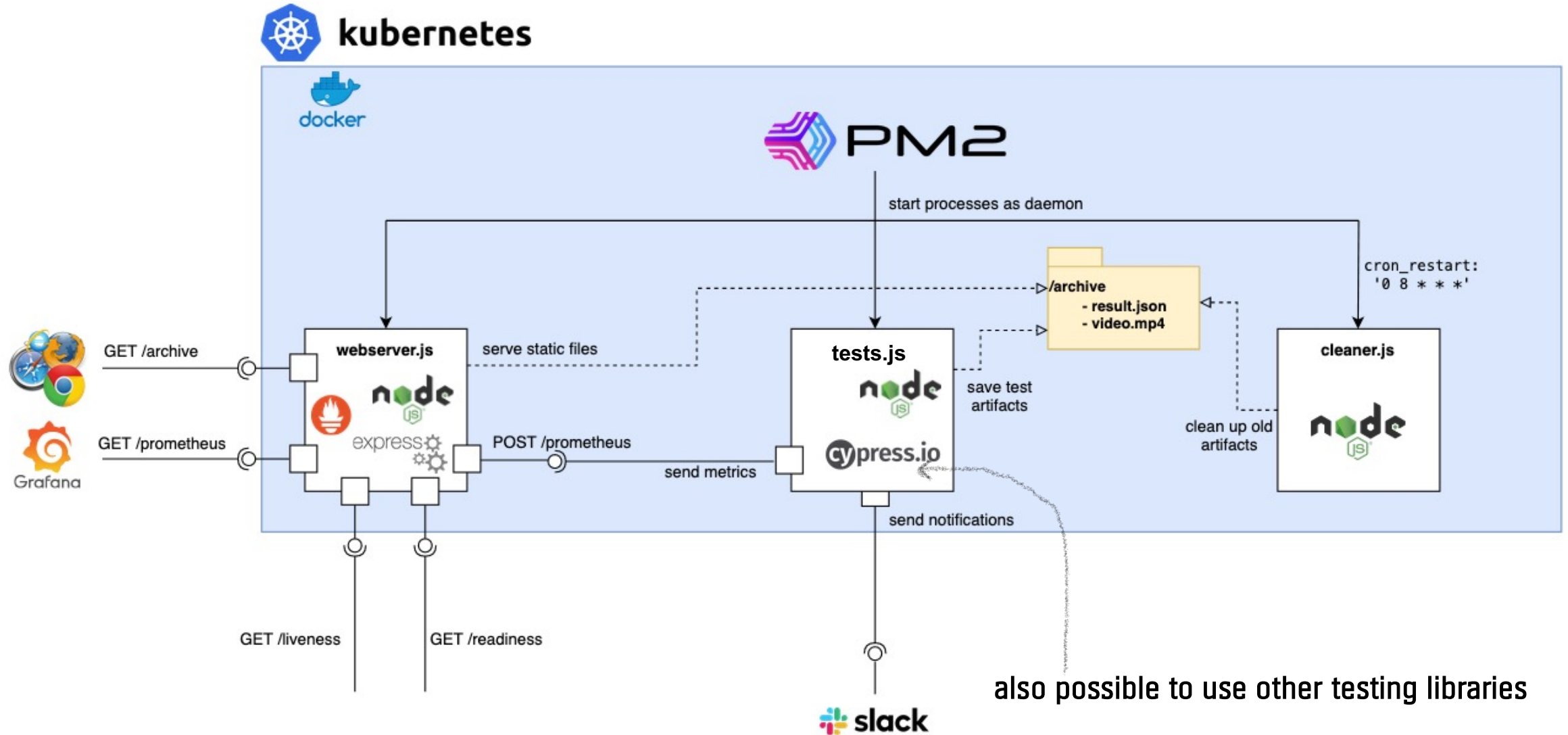
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Solution: The Big Picture



Solution: The Big Picture



Solution: Daemon process manager

Daemon process manager for Node.js  PM2

Dockerfile

```
FROM cypress/browsers:node-18.16.0-chrome-112.0.5615.121-1-ff-112.0.1-edge-112.0.1722.48-1
RUN apt-get -y update; apt-get -y install curl tzdata tini

# set the correct timezone for correct timestamps in logging
ENV TZ=Europe/Berlin
RUN ln -snf /usr/share/zoneinfo/$TZ /etc/localtime && dpkg-reconfigure -f noninteractive tzdata

RUN mkdir /opt/prj/
WORKDIR /opt/prj/
# only copy "node_modules", if it is present
COPY package.json package-lock.json *node_modules /opt/prj/

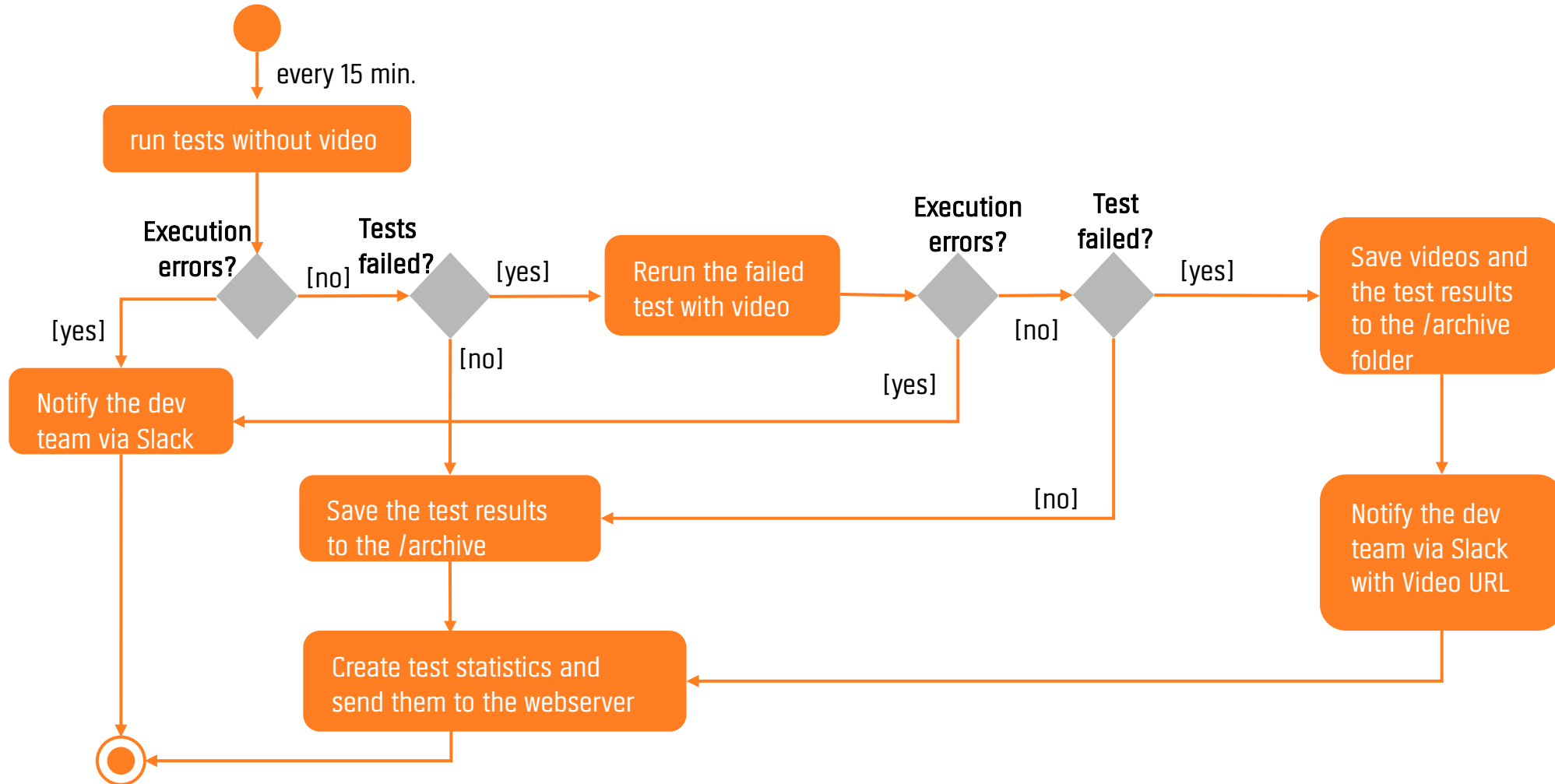
# check Cypress binary path and cached versions
RUN npm install
RUN ./node_modules/.bin/cypress verify

COPY . /opt/prj/
ENTRYPOINT [ "/usr/bin/tini", "-g", "--", "/opt/prj/node_modules/.bin/pm2-runtime", "ecosystem.config.js" ]
```

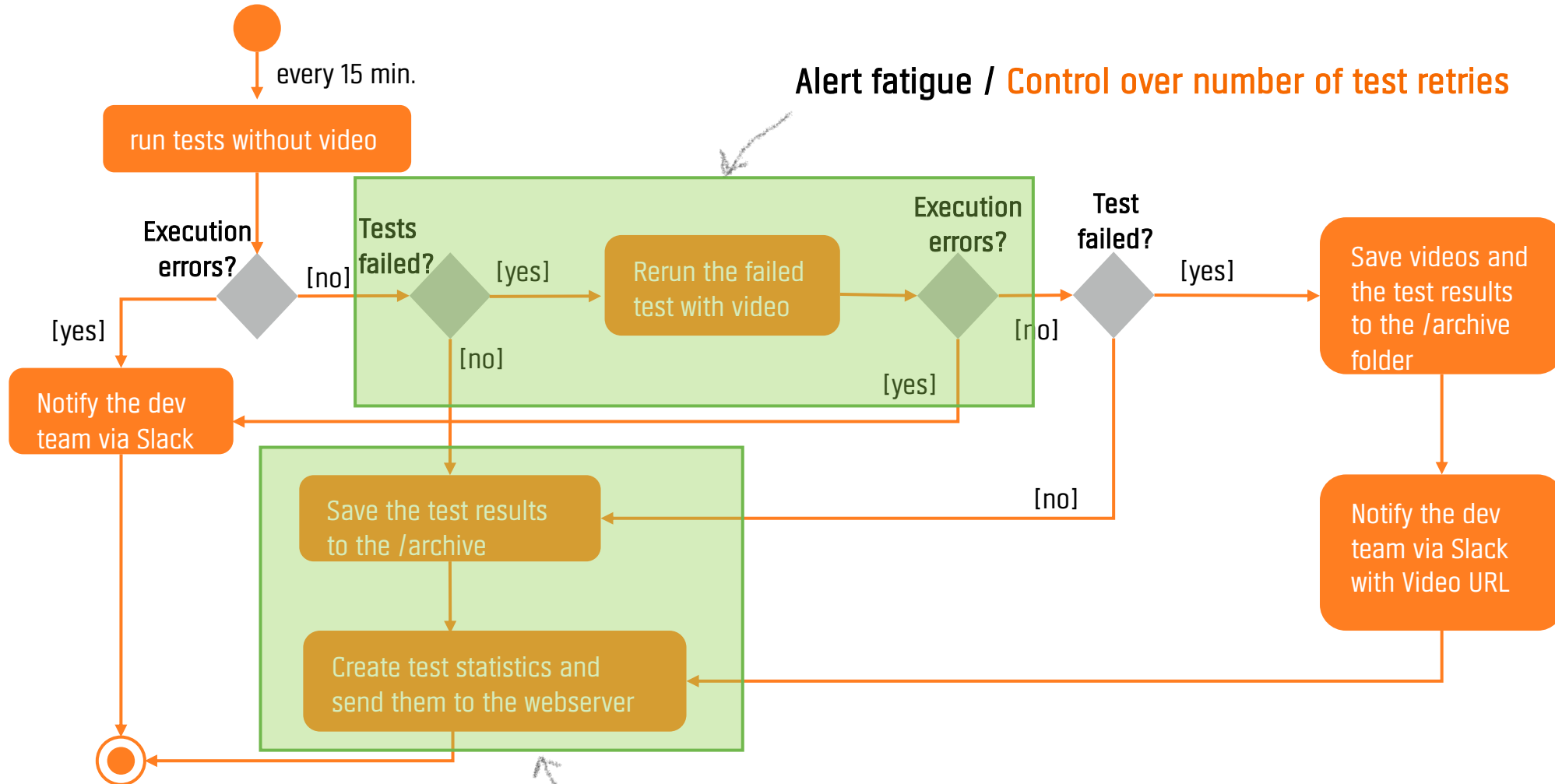
ecosystem.config.js

```
module.exports = [{
  script: 'webserver.js',
  name: 'webserver'
}, {
  script: 'cypress-tests.js',
  name: 'cypress',
  // don't start automatically,
  // so kubernetes can restart
  // the whole pod by itself
  autorestart: false
}, {
  script: 'cleaner.js',
  name: 'cleaner',
  cron_restart: '0 8 * * *',
  autorestart: false
}]
```

Solution: Test execution (activity diagram)



Solution: Test execution



Alert fatigue / Control over number of test retries

Storing historical monitoring data / Statistics and Analysis

Solution: Prometheus custom metrics

```
const prom = require('prom-client');
const register = new prom.Registry();

const e2eTimeGauge = new prom.Gauge({
  name: e2e_time',
  help: 'Duration of Cypress Tests in milliseconds',
  labelNames: ['name'],
});

const e2eResultGauge = new prom.Gauge({
  name: e2e_result',
  help: 'Results of Cypress Tests',
  labelNames: ['name'],
});

...
e2eTimeGauge.labels({name: metric.name}).set(metric.time);
e2eResultGauge.labels({name: metric.name}).set(metric.result);
...

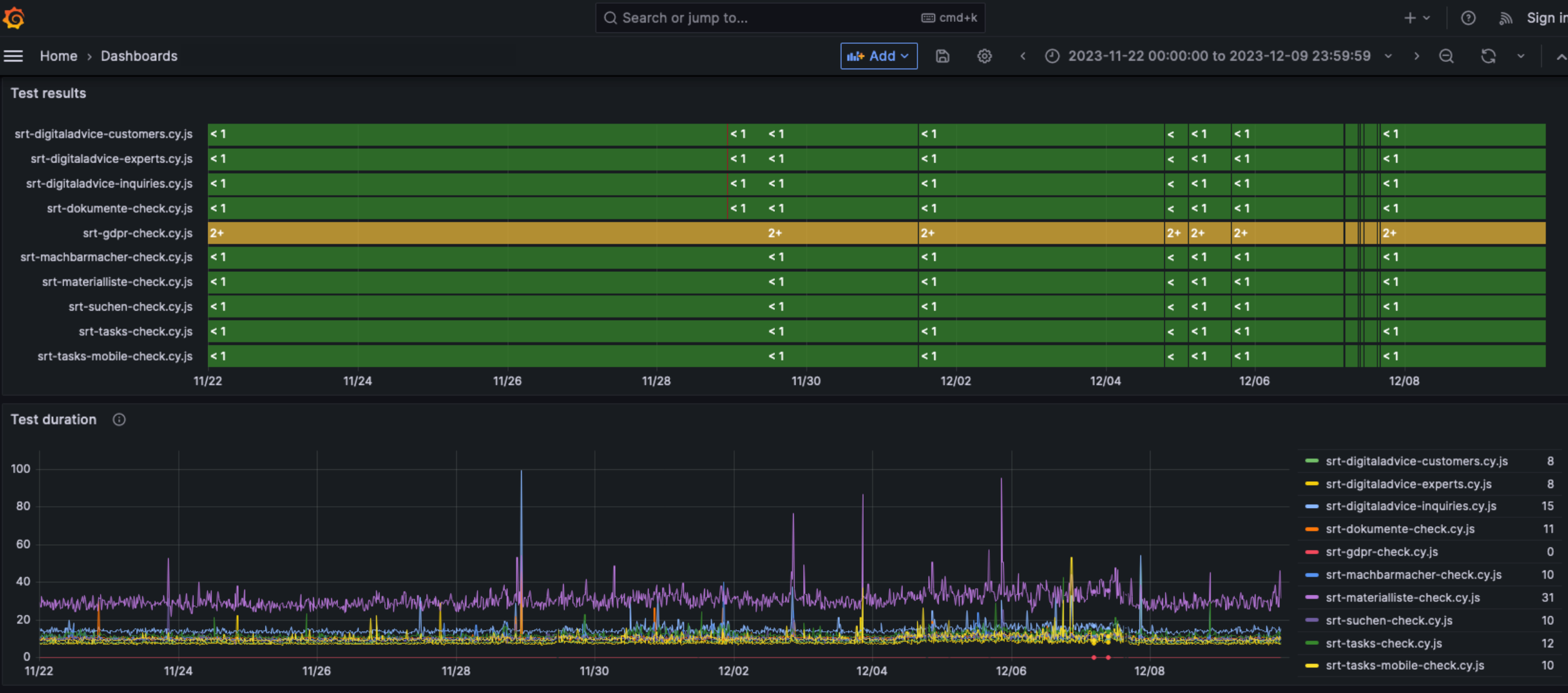
app.get('/prometheus', async (req, res) => {
  res.setHeader('Content-Type', register.contentType);
  res.send(await register.metrics());
});
```



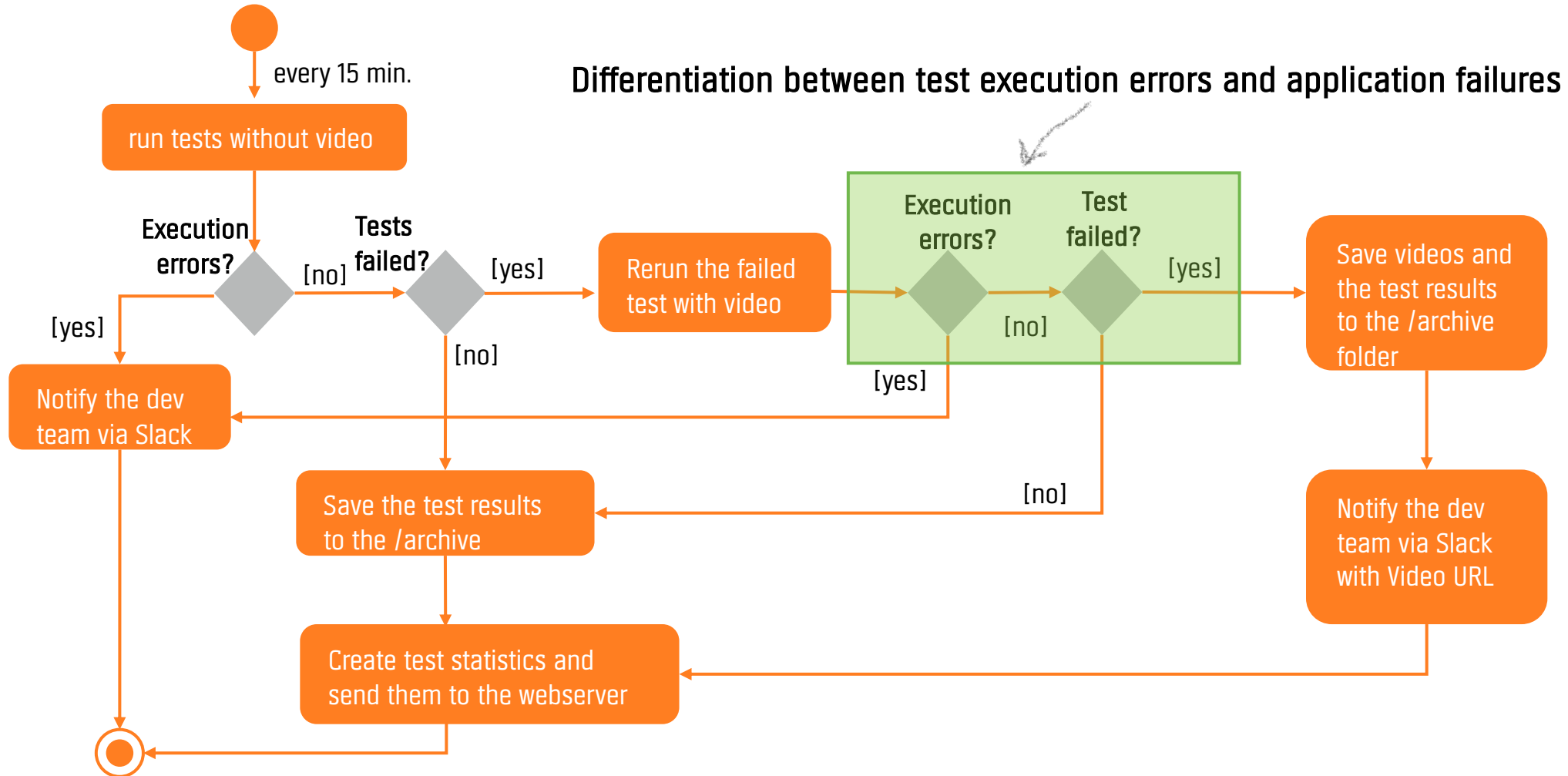
Prometheus

<https://prometheus.io/docs/instrumenting/clientlibs/>

Grafana Dashboard



Solution: Test execution (activity diagram)



Slack Notifications:



SRT-Notifier APP 20:51



Problem detected in the Use Case XYZ in stage!

Details: test has failed. Please see [Videos and Logs](#).



Failed tests



SRT-Notifier APP 06:31



The e2e-tests has been killed and will be automatically restarted by Kubernetes
The Test Runner unexpectedly exited via a exit event with signal SIGKILL

Please search Cypress documentation for possible solutions:

<https://on.cypress.io>

Check if there is a GitHub issue describing this crash:

<https://github.com/cypress-io/cypress/issues>

Consider opening a new issue.

Platform: linux-x64 (Debian - 11.6)

Cypress Version: 12.16.0



Execution errors

Archive with test results and videos

~ / archive /

2023-12-09T17:50:05.815Z
2023-12-09T17:51:09.140Z
2023-12-09T17:52:22.521Z
2023-12-09T17:53:23.857Z
2023-12-09T18:09:21.107Z
2023-12-09T18:10:03.381Z

2023-12-09T17:50:21.804Z
2023-12-09T17:51:15.531Z
2023-12-09T17:52:43.689Z
2023-12-09T18:08:40.656Z
2023-12-09T18:09:39.280Z
2023-12-09T18:10:49.979Z

The file name is the test start time

~ / archive / 2023-12-09T17:50:05.815Z /



result.json

Solution: The Webserver - liveness and readiness Check



```
app.get('/liveness', (req :... , res :Response<ResBody, LocalsObj> ) => {  
  isProcessOnline(CYPRESS_PROCESS).then(result => {  
    if (!result) {  
      logger.error(`Liveness check failed. Missing process: ${CYPRESS_PROCESS}`);  
      res.status( code: 500).send( body: `Process is missing: ${JSON.stringify(CYPRESS_PROCESS)}`);  
    } else {  
      logger.info(`All processes live`);  
      res.status( code: 200).send( body: 'All processes live');  
    }  
  }).catch(error => {  
    logger.error(`Error fetching /liveness: ${error}`);  
    res.status( code: 500).send( body: `Error fetching /liveness: ${error}`);  
  });  
});
```

automatic restarts if e2e
process died

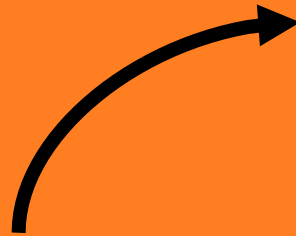
The logs are stored for 30 days

```
function isProcessOnline(expectedProcess) {  
  return new Promise( executor: function (resolve, reject) {  
    pm2.list( errback: function (err :Error , processes :ProcessDescription[] ) {  
      if (err){  
        reject(err);  
      }  
      const liveProcesses = processes.filter(process => process.pid > 0).map(process => process.name);  
      resolve(liveProcesses.includes(expectedProcess));  
    });  
  });  
}
```

additional protection:
Grafana Alerting, if no
"test start" logs for
20 Minutes

QUESTIONS?

Let's connect on LinkedIn



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